

GOOGLE DATA ANALYTICS CAPSTONE

Sleep & Activity: What the Data Shows

A marketing analytics case study for Bellabeat, using FitBit Fitness Tracker Data to inform recovery-focused product messaging.

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The business task

Bellabeat's marketing analytics team was asked to analyze non-Bellabeat smart device data to uncover usage trends, then apply those insights to one Bellabeat product to guide marketing strategy.

Requested by Urška Sršen (Cofounder & Chief Creative Officer) and Sando Mur (Cofounder).

Working question

Does sleep duration relate to physical activity levels — and can it inform a recovery-focused marketing angle?

Deliverable

High-level marketing recommendations, grounded in data.

The data

FitBit Fitness Tracker Data

Kaggle, CC0 Public Domain, made available through Mobius. Personal tracker data from 30 consenting Fitbit users.

Files used: dailyActivity_merged, sleepDay_merged

Known limitations (ROCCC)

- Small sample — 34 unique users across files
- Self-selected sample, via Amazon Mechanical Turk
- Collected in 2016 — dated relative to current usage
- Inconsistent daily logging by users

Two lenses on the same question

Lens 1 — Day to day

Matched each sleep record to the following day's activity, by user and date, to test a direct predictive link.

Lens 2 — Person level

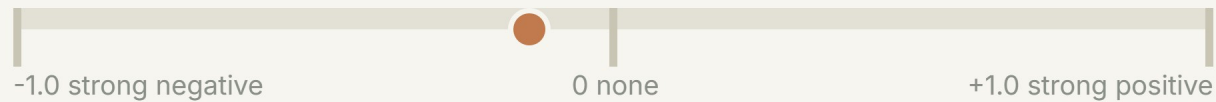
Segmented users into four activity levels (CDC step thresholds) and compared each segment's average sleep.

Tool Google Sheets — pivot tables, INDEX/MATCH, and CORREL(), with every cleaning step documented in a changelog.

Sleep alone doesn't predict next-day activity

-0.14

CORRELATION COEFFICIENT



A correlation this close to zero means, day to day, one night's sleep is not a meaningful predictor of the next day's step count.

But active users, as a group, sleep less



Average sleep duration per user, grouped by activity segment (CDC step thresholds). 34 users total, 7–10 per segment.

Noise at the daily level, a pattern at the person level

Day to day

One night's sleep is drowned out by workload, stress, and one-off rest days — too noisy to predict the next day alone.

Person level

Averaging away that daily noise reveals a stable trait: habitually active people sleep less, as a consistent group pattern.

Weak individual-level correlation does not rule out a real group-level pattern — they answer different questions.

Recommendation

Introduce targeted sleep-recovery messaging for Bellabeat's most active users — positioning the product as supporting balance between activity and rest, not just tracking steps.

1

Target recovery content and reminders at highly active users specifically, not all users generically.

2

Position Time and Leaf's sleep-tracking features as tools for balancing activity with recovery.

3

Validate with a larger sample and Bellabeat's own device data before scaling the campaign.

LIMITATIONS & NEXT STEPS

Directional, not definitive

- 34 total users, 7–10 per segment — a real but small sample
- Self-selected participants may not represent all users
- 2016 data may not reflect current activity habits

Next

Re-test this pattern with Bellabeat's own first-party device data at scale before committing marketing budget.

Thank you

Bellabeat Marketing Analytics — Sleep & Activity Analysis